Semantic Integration of Textual Data in Official Statistics:

From Deterministic Coding to Language Models

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Official statistics rely on structured classifications

- Official statistics transform real-world phenomena into structured categories.
- Classifications such as **NACE**, **ISCO** provide a shared statistical language.
- These systems are essential for **consistency, comparability, and integration**.

But integrating new and rapidly evolving data sources into these structured systems is becoming an increasingly important challenge.

Textual data in Official Statistics

Many emerging data sources used in official statistics contain **unstructured textual information**.

Examples include:

- **Administrative data** – demographic, economic, fiscal records
- **Web data** – information about firms' products and services.
- **Public online data** – sources such as patents

These sources contain valuable information but are **not directly compatible with structured statistical classifications**.

Traditional automated coding systems

Traditional systems assign classification codes to textual descriptions using **deterministic rules**.

Typical components include:

- **Dictionaries** derived from classification descriptions
- **Parsing and normalization rules**
- **Matching strategies** based on word overlap

Examples used in official statistics include systems such as **ACTR** and **CIRCE**.

These systems match words, but they do not capture meaning.

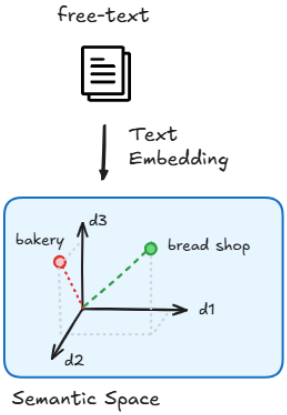
Matching meaning with AI

Modern AI models represent text as **numerical vectors**, called **embeddings**.

In this representation:

- texts with similar meanings are located close to each other
- similarity can be measured using **distance metrics** (e.g. cosine similarity)

This allows automated coding systems to match **meanings** rather than individual **words**



Lexical vs. semantic representations

01.11 Growing of cereals, other than rice, leguminous crops and oil seeds

Deterministic representation

Pre-processing:
build a lexical dictionary

[Grow, cereal, rice, legume, crop, oil, seed]

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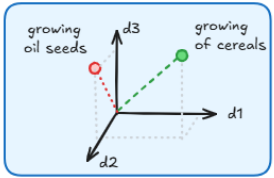
Weight	n-grams
1.0	grow&cereal
0.25	grow&crop
0.5	grow&seed
...	

Semantic representation

Growing of cereals
Growing of leguminous crops
Growing of oil seeds

Sentence Embedding



Assessing Reliability of AI-based coding

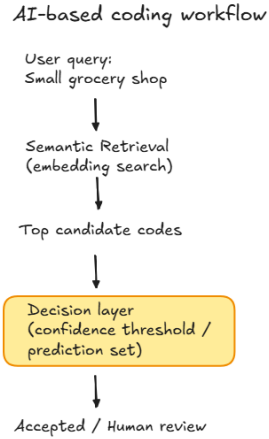
Semantic retrieval generates candidate codes.

How can we assess whether **the prediction can be considered reliable**?

Reliability can be assessed through:

- confidence thresholds
- prediction sets
- uncertainty quantification

Developing these methods requires **new statistical and computational skills** within statistical institutions.



AI in official statistics: opportunities and responsibilities

- **New data integration opportunities.** Modern AI methods enable statistical institutions to integrate large volumes of textual and unstructured data into established classification systems.
- **Statistical reliability remains essential.** AI predictions must be supported by quality assessment, uncertainty quantification, and robust statistical validation procedures.
- **New skills for statistical institutions.** AI is transforming how statistical research is conducted, requiring new methods, tools, and stronger AI literacy across statistical institutions.

Istat

Q&A

Thank you for listening!

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